

Defense Portfolios for Synthetic Hospital Ransomware Scenarios: A Partially Calibrated Multi-objective Decision Analysis with Full-Space Confirmation

Ashish Agrawal¹ Mukil Dharanidharan¹ Naman Upadhyay¹

¹Independent Student Researchers, Naperville, Illinois, USA

Abstract Placeholder section.

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1 Introduction

Ransomware can interrupt care, displace demand to neighboring hospitals, and extend recovery far beyond the restoration of a server. Among 374 publicly reported attacks on U.S. health care delivery organizations from 2016 through 2021, 44.4% mentioned care disruption and 41.7% mentioned electronic-system downtime [1]. WannaCry was associated with lost appointments and reduced activity at affected National Health Service trusts [2]. A month-long U.S. attack was followed by increased demand, longer waits, and stroke-care delays at neighboring emergency departments [3], and related work reported clinical changes at nearby, unaffected hospitals [4, 5]. A first-party incident review documents an attack in which access preceded detonation by roughly eight weeks while restoration unfolded over months [6].

These studies establish that cyber incidents become operational and regional events. None reveals the counterfactual: what the same incident would have done under a different defense portfolio. That counterfactual is what a hospital needs, and it is not publicly observable.

Simulation can supply one by replaying a single latent incident under different defenses. Doing so honestly raises the two problems this paper is about.

The first is identification. A model of this kind contains dozens of internal coefficients — per-edge transmission, patch effectiveness, isolation success, restoration throughput — and public evidence identifies almost none of them. Public data constrain incident recovery classes, multi-facility breadth, entry-vector families, facility context, control *adoption*, backup outcomes, and patient-flow effects. They do not constrain a segmentation multiplier or a per-step isolation probability. A model that fills every parameter slot with a citation is not better calibrated than one that does not; it is less honest about the same uncertainty. We therefore classify every quantity by what identifies it, and report the unidentified ones as declared assumptions rather than estimates.

The second is the decision. Segmentation, patching, detection, isolation, identity controls, and backup protection impose different costs and workflow burdens. A portfolio that minimizes average disruption may be unaf-

fordable, behave poorly in the tail, or depend on aggressive isolation that removes healthy systems from service. Multi-objective optimization preserves non-dominated alternatives instead of hiding preference choices inside one score [7, 8], and cybersecurity investment research has long treated defense selection as a constrained decision problem rather than a one-dimensional ranking [9, 10].

1.1 This paper corrects its own predecessor

An earlier version of this study reported a six-objective frontier with a frozen holdout. An audit found defects severe enough that its results could not be repaired by re-analysis. We report them rather than quietly superseding them.

- The sustained-outage endpoint was defined in prose as requiring at least four clinical services and computed in code as requiring one. It was one of six objectives, so the inversion propagated into the frontier, the stability estimates, the finalist rule, and every reported outage percentage. A 70-test suite[†] passed both before and after the definition was corrected, because no test asserted the manuscript’s definition of anything.
- Per-service outage durations were never written to raw output, so the endpoint could not be recomputed at any other service count from the published data. That is why the present study is a fresh run rather than a re-analysis.
- Capacity profiles were overwritten by portfolios at zero cost. Every candidate declared both a segmentation and a backup architecture, the portfolio value always won, and the weakest rung of each ladder was priced at nothing — so a strong hospital could “buy” flat segmentation and connected backups for zero points. Four of the seven frozen high-capacity finalists[†] were such free downgrades.
- The holdout frontier ranged over 57 discovery-selected finalists[†] rather than the resolved candidate space, and was presented without a candidate-set qualifier.
- The “frozen” holdout protocol was rewritten by its own runner on every invocation and contained no hash, commit, or timestamp. Its claim to have been frozen prospectively cannot be checked from the public

record, and we do not repeat it.

Figures marked [†] come from the forensic audit of the superseded version, which is committed alongside this work; they describe an artifact that no longer exists and so cannot be regenerated from current tables.

We state this history because a reader is entitled to know which numbers were withdrawn and why, and because the corrections determined what the present design had to be. The superseded data are retained in the repository, marked as uncitable.

1.2 Contribution

The question is narrow:

Which defense portfolios remain Pareto-efficient across mean disruption, tail disruption, sustained clinical outage, non-recovery, normalized cost, and normalized operational burden, when every resolved candidate is evaluated on a fresh paired scenario bank under a protocol frozen beforehand?

Four contributions. First, an explicit parameter-identification framework that classifies every reported quantity and refuses to present declared assumptions as evidence. Second, a corrected model in which environmental capacity and purchased controls are genuinely separate. Third, a *full-space* confirmatory frontier, which lets us measure directly how many candidates appear efficient only because the candidate that dominates them was never evaluated. Fourth, a reproducibility apparatus in which every number in this manuscript is generated from a committed table, the confirmatory protocol is content-addressed and cannot be rewritten, and a claim audit fails the build when prose and artifacts disagree.

This is a computational decision analysis. It is not an empirical estimate of any hospital's attack probability, and it is not evidence that any control achieves any effect in a real hospital.

2 Evidence and parameter identification

Every quantity carries an identification class, declared in a machine-readable registry that the analysis code, the tables, and the claim audit all read. The classes are *evidence-constrained* (matched public evidence on a matching population), *externally validated* (an observable output compared against a frozen benchmark), *declared assumption* (a modeling choice, with no evidence claim), *model-internal* (no real-world referent), and *normalized scenario unit* (explicitly not dollars, staff time, or risk).

2.1 What public evidence can constrain

Incident-level disruption and recovery classes are observable: the 2016–2021 cohort documented care disruption in 166 of 374 attacks, electronic-system downtime in 156, scheduled-care delay or cancellation in 38, and ambulance diversion in 16, with durations binned from under one week to over four weeks and a further 67 events of known disruption but unknown duration [1].

Attacked-hospital activity is observable: volume declines of roughly 17–24% in the first week with recovery over about three weeks [11], alongside changes at nearby unaffected hospitals [3, 5]. Facility context is observable through public cost-report data [12]. Control *adoption* is partially observable through sector survey work [13].

These support scenario construction and external validation targets. None is a control-effectiveness estimate.

2.2 What public evidence cannot identify

No public source provides a representative internal hospital network topology, a per-edge transmission probability, a universal patch effectiveness, a segmentation multiplier, a per-step isolation success probability, an identity-compromise spread multiplier, a traversal probability into isolated backups, a node restoration throughput, clinical service functional thresholds, false-positive isolation incidence, or control-specific capital and workflow costs. These are **non-identifiable** from public data and are reported as declared assumptions throughout.

Two sources are commonly misread, and we name the misreadings because we were at risk of both. Vendor surveys report entry vectors among self-selected respondents across many countries and organization sizes, with categories that are neither exhaustive nor mutually exclusive and that moved substantially between adjacent years [14–16]. They justify named scenarios, not a prevalence. Observational authentication research estimates a large relative reduction in compromise *at covered authentication events* [17]; it does not license multiplying whole-network ransomware probability by that figure, because it says nothing about unenrolled accounts, service accounts, legacy protocols, token theft, or compromised administrators.

Guidance documents [18–20] justify mechanisms and scenario structure. They are not cited as estimating any coefficient.

2.3 Assumptions this model makes that it should not

Three parameters are used in ways this framework does not sanction. We report them as defects rather than defending them.

- **Patch effectiveness** is a single scalar applied to every patched node against every pathway, including credential-origin and vendor-origin scenarios where patching is mechanically irrelevant. It should be exploit-specific susceptibility removal, or an explicitly broad uncertainty distribution.
- **Identity compromise** multiplies every edge in the network rather than the eligible authentication pathways.
- **Isolated backups** are unreachable by construction: the traversal multiplier into an isolated backup zone is exactly zero, so isolated backups cannot fail by any modeled mechanism. Survey evidence reports attempted backup compromise in the large majority of healthcare victims and success in a substantial minority of attempts [15]. A hard zero is not a defensible representation of any real architecture, and it makes

protected backups a deterministic win in the objective space. A deliberately failing test keeps this visible in every test run so it cannot be forgotten between versions.

3 Model specification

The full specification is a versioned document in the repository; the endpoint definitions are a Python module that the analysis code, the tables, and the claim audit all read. What follows is a summary.

3.1 Structure and dynamics

The model contains no malware, exploit code, scanning, or connection to real infrastructure. A generated directed graph represents permitted access paths among workstations, EHR, laboratory, pharmacy, imaging, medical-device, administrative, identity, backup, and internet-facing or vendor functions. Nodes move among healthy, compromised, detected, isolated, restoring, and restored states. A node is functional when it is neither compromised nor isolated; false-positive isolation removes healthy nodes from service for a bounded period, which is a modeled operational cost of aggressive automated response rather than an error.

Seven services depend on core and supporting nodes. A service is available at a step when at least 60% of its supporting nodes are functional. Service availability is **binary at each step**, so the model cannot represent degraded operation, downtime procedures, or manual workarounds. The 60% fraction is a declared assumption with no clinical validation.

Each step is 5 modeled minutes and the horizon is 72 hours. Within a step, spread precedes detection, detection precedes isolation, false positives are resolved, restoration proceeds, and service availability is scored.

Structural assumption S1. Restoration begins only once the estate is contained — once no compromised node remains un-isolated. Real incident response restores in parallel with containment and prioritises clinical systems. This couples the detection and isolation controls to the recovery endpoint through a modeling choice rather than a mechanism. It was undocumented in the previous version; it is named here and varied in the structural sensitivity analysis.

3.2 Capacity is exogenous; controls are additive

A *capacity profile* is the environment a decision-maker inherits: starting patch coverage, detection delay, isolation success, legacy fraction, baseline segmentation, baseline backup architecture, and a normalized budget. The three profile names are neutral scenario labels, not observed hospital classes.

A *portfolio* is a bundle of purchased upgrades layered on top of a profile: segmentation (flat, basic, least-privilege), patch coverage (zero to three ladder rungs), improved detection, rapid same-step isolation, backup architecture (connected, periodic, isolated), and identity controls. Their Cartesian product gives 288 enumerated portfolios.

Two rules make the separation real, and both were absent before.

Upgrade-only precedence. For every posture ladder, the effective setting is the stronger of the profile baseline and the portfolio target. A portfolio can raise a profile's posture, never lower it.

Increment pricing. A portfolio is priced on the rungs it actually buys relative to its profile's baseline, on both the cost and the burden tables. A downgrade costs zero rather than refunding, and a profile is never charged for a rung it already has.

Together these guarantee that exactly one candidate per profile is free — the profile's own untouched posture — which the test suite asserts. They also reduce the resolved candidate space, because a strong profile has fewer decisions left: 288, 96, and 16 behaviorally distinct candidates in the resource-constrained, intermediate-capacity, and high-capacity profiles respectively. The high-capacity count is small because that profile already owns least-privilege segmentation and isolated backups. That is the honest size of its decision space.

3.3 Endpoints

Table 1 gives the six minimized objectives. All are reported separately; no hidden utility function combines them. Defensive isolation node-hours per node is retained as a diagnostic and is not a seventh dominance dimension.

3.4 The sustained-outage endpoint, and a withdrawn justification

This endpoint caused the previous version's central defect, so we set the decision out in full.

The rule is *k-of-n*: a trial qualifies when at least *k* of the four clinical services each sustain a continuous outage of more than 2 modeled hours. The comparison is strict, and the runs are *not* required to be concurrent — each service is scored on its own longest outage over the horizon, because per-step service states are not retained for every trial. We use the weaker reading and say so wherever the endpoint appears.

An earlier draft of the rebuilt specification set $k = 2$, arguing that $k = 1$ would saturate and $k = 4$ would be too rare to estimate. **The data do not support that argument and we withdrew it.** Across 12,000 paired discovery trials the number of qualifying clinical services is almost perfectly bimodal: 4,189 trials had none and 7,269 had all four, with only 542 trials (4.5%) anywhere in between. Candidate-level outage probabilities therefore differ by about three percentage points between $k = 1$ and $k = 4$, and the interquartile spread across candidates is identical to three decimal places.

Because the choice of *k* costs nothing in estimability and buys nothing in discrimination, it is settled on construct grounds alone, and we take the strictest reading: $k = 4$,

Table 1: Decision objectives with the identification class of each. Only simulated outcomes share units with the model. Cost and burden are normalized scenario points.

Objective	Definition	Identification	Interpretation boundary
Mean disruption	Mean weighted service-hours lost across the seven modeled services, each binary at each step	Model-internal	Expected service unavailability within the synthetic model; not patient harm
Tail disruption	Conditional mean weighted service-hours lost among trials at or above the empirical 90th percentile	Model-internal	Severity of the worst simulated decile under the model's own scenario bank
Sustained clinical outage	Fraction of trials in which at least k of the four clinical services each experienced a continuous unavailability run of more than 2 modeled hours, with $k = 4$ primary and every k reported	Declared assumption	A model indicator of multi-service clinical disruption; not an incident probability and not a clinical threshold
Non-recovery	Fraction of trials in which weighted clinical availability had not returned to full at the final simulated step of the declared horizon	Model-internal	Right-censored technical recovery within the horizon; not organizational recovery
Implementation cost	Sum of declared control cost points for the increment a portfolio adds to its profile's exogenous baseline	Normalized unit	A relative scenario weight; not dollars and not procurement evidence
Operational burden	Sum of declared workflow and implementation burden points for that same increment	Normalized unit	A declared scenario preference; not measured staff time or alert volume

the definition the previous manuscript stated and the code did not implement. Choosing a looser k would have moved the definition closer to where the defect happened to sit, with no evidence for it. Every results table reports all four values of k , so a reader who prefers a different one reads it off the same table.

The bimodality is itself a model artifact. Binary per-step service availability over a shared network structure produces all-or-nothing clinical outcomes, whereas real incidents show partial, degraded, and staggered service loss. This limits what the endpoint can represent regardless of k , and it is a more serious caveat than the choice of k ever was.

3.5 What the recovery endpoint cannot do

Non-recovery is right-censored technical recovery at 72 hours. The restoration rate implies that a few-hundred-node estate is restored within hours once contained, whereas the public incident record has a substantial weeks-to-months tail. **A 72-hour horizon with this restoration rate cannot express the observed tail at all**, so non-recovery is an artifact of the horizon rather than a recovery estimate, and we report it as one. Separating acute service availability, minimum viable operations, and full organizational recovery into distinct endpoints over 72 hours, 21 days, and 90 days, with a 180-day tail scenario, is the next required piece of work and is not attempted here.

4 Study design

4.1 Stage separation

Development, discovery, confirmation, and external validation are separated, and the separation is enforced

by artifacts rather than by discipline. Discovery is exploratory: it debugs the specification, estimates Monte Carlo precision, and generates candidate labels. It is labeled non-confirmatory in every output. The confirmatory stage reuses no discovery scenario, and individual cells are never selectively rerun.

4.2 Precision, not round numbers

The scenario count comes from a pilot precision analysis rather than a declaration. For each profile and objective we estimate the per-scenario standard deviation, take the resolution target to be one tenth of the interquartile range of candidate means — the separation the frontier must actually distinguish — and solve for the required scenario count, using the observed within-scenario correlation to obtain the paired requirement that governs a frontier comparison.

The requirement varies by more than a factor of four across profiles. Since pairing is within a profile, nothing requires a common count, so scenarios are allocated per profile: , , and scenarios per candidate, giving confirmatory executions. Where a count still falls below a requirement we say so and treat the affected objective as under-resolved rather than established.

4.3 Full-space confirmation

The confirmatory frontier ranges over every resolved candidate in each profile.

This is the design change that most distinguishes this study from its predecessor. Evaluating only discovery-selected finalists makes the reported frontier conditional on discovery in a way the reader cannot see: a candidate that was mediocre in discovery and excellent on fresh

scenarios is structurally unable to appear, and a small restricted set is close to arithmetically guaranteed to be mostly non-dominated in six dimensions. Measured throughput made the honest version affordable, so we compute the true frontier rather than rename the restricted one.

The previously frozen finalist rule is retained, but it now labels rather than restricts. Because the finalists are a subset of what we evaluate, the finalist-only frontier is a *view of the same data*, and we report directly how many candidates appear efficient only because the candidate that dominates them was never evaluated. That quantity was unmeasurable before.

4.4 Protocol freezing, and an honest statement about the past

The confirmatory protocol is a content-addressed artifact. Its own SHA-256 is computed over its canonical form and stored inside it; the freezing routine refuses to overwrite an existing protocol; loading recomputes the digest and fails if the file has been edited. The runner will not start without a protocol that verifies, refuses to run if the resolved candidate space has changed since freezing, and stamps the digest into every raw row and into the run manifest. The protocol used here is , frozen and committed before any confirmatory output existed.

We do not claim the previous study's freeze was prospective. It cannot be established from the public record, so that analysis is described as retrospective. Only protocols frozen under the mechanism above support a prospective claim, and only from their freeze timestamp forward.

4.5 Pairing and uncertainty

All candidates within a profile replay one scenario bank: the same topology, entry point, patch draws, and counter-style event-level random fields. Controls may delete or modify paths but cannot receive more favorable latent randomness. This is a common-random-numbers design [21].

We distinguish aleatory event variation, parameter uncertainty, structural uncertainty, evidence uncertainty, and Monte Carlo error, and report the last explicitly. Every stochastic objective carries a standard error, and we report the number of candidate pairs separated by less than one combined standard error on every stochastic objective — pairs the scenario bank cannot reliably order, whatever the dominance test says about their point estimates. Frontier stability uses paired scenario bootstraps that resample whole scenario identifiers, preserving cross-portfolio correlation.

5 Results

Placeholder section.

6 Discussion

Placeholder section.

6.1 Validity, separated

Conflating these is how a computational study acquires more credibility than it has earned, so they are reported separately and the weakest is named as the weakest.

Model validity. The mechanism is an abstract epidemic process on a declared graph. It is defensible as a decision-analysis scaffold and it is not validated against internal hospital telemetry, because no representative public telemetry exists. Structural assumption S1 is varied rather than asserted; the remaining structure is not.

Data validity. The archived public sources are hash-verified and every count the registry records re-derives from the file. One source is a normalized export of a browser preview because the original requires sign-in; the recorded hash makes any substitution detectable, and replacing it with the depositor's original is a pre-submission task.

Construct validity. This is the weakest link, and it is weaker than the previous version admitted. Binary per-step service availability cannot represent degraded operation, downtime procedures, or manual workarounds, and that limitation is not merely a caveat — it produces the near-binary clinical outcome distribution reported above, which in turn is why the choice of k makes almost no difference. Weighted service-hours are not patient harm. Normalized cost points are not procurement.

External validity. Topology, transition hazards, service dependencies, control effects, cost, and burden are not jointly observed in any real hospital. Nothing here estimates what a deployed control would do. The frozen benchmark suite is the check on this, and most of it cannot be answered by a model of this shape.

Decision relevance. The outputs support tabletop exploration of where a decision hinges on budget or preference. They do not support procurement, and a portfolio's position on a frontier is not a recommendation.

Computational reproducibility. This is the strongest link, and it is worth being precise about what that buys. Every number above is generated from a committed table by script; every input and output is hashed in a manifest that fails rather than warns; the confirmatory protocol is content-addressed and was committed before any confirmatory output existed; and a claim audit fails the build when prose and artifacts disagree. None of that makes a conclusion true. A perfectly reproducible pipeline reproduces a wrong answer exactly. What it does is make the wrongness locatable.

6.2 Limitations

Reporting bias. Public ransomware incident data record what was publicly reported and linkable. A zero operational flag may mean "not reported" rather than "did not occur", and 67 of 374 events in the incident cohort have known disruption of unknown duration — missingness that is plausibly informative, since longer or more severe incidents may be less likely to have a cleanly reported end date.

Vendor surveys. Entry-vector and backup figures come from self-reported surveys spanning many countries and organization sizes, with non-exclusive categories that moved substantially between adjacent years. They are graded below administrative and peer-reviewed hospital evidence throughout, and they inform scenario names rather than parameter values.

International transferability. The most detailed first-party incident review available is from a non-U.S. health system [6]. It supports mechanism, not magnitude.

Missing denominators. The exploited-vulnerability catalog has no hospital exposure denominator, so its ransomware-tagged share is not a base rate and is never used as one.

Interval censoring. Recovery evidence arrives in duration bins with a substantial unknown-duration group. Any recovery-class model fitted to it must test the missingness assumption rather than dropping those events.

Absent internal telemetry. There is no representative public measurement of hospital internal network structure. The generated graph is a declared structure, not a measured one, and node counts are not a function of beds.

Elicited thresholds. The service-functional fraction, the outage duration, and the service count are declared choices without clinical validation. Calling any of them a clinical threshold would be false.

Cost uncertainty. Normalized points omit procurement, integration, training, maintenance, alert workload, and constraints specific to clinical technology. They are ordinal scenario weights and nothing more.

Adaptive adversaries. The model has none. An attacker who responds to a deployed defense is not represented, so every control effect here is an effect against a static process.

Cross-hospital dependence. The simulator runs one facility. The incident record contains multi-hospital events and documented regional spillover [3, 5]. This is a failure boundary, reported as one rather than absorbed.

Non-causal control effects. No number in this paper is a causal estimate of a control's effect. Every control effect is a conditional model output under declared assumptions.

6.3 What would actually improve this

Not more simulations. The binding constraint is identification, and it is relieved only by data this project does not have: de-identified telemetry from several hospitals, with denominators — asset inventories with patchability and segment membership; vulnerability findings with first-seen, patch-available, and remediated dates; authentication events with coverage, method, and privileged-account class; high-confidence incident timestamps from initial access through recovery milestones; isolation actions with adjudication, release time, and clinical impact; backup architecture with restore-test outcomes and clean-recovery-point availability; and service-level availability with staffing and patient volume. A hierarchical model with hospital-level random effects and explicit measurement error would then be estimable.

Failing that, structured expert elicitation with calibration questions and pooled uncertainty is preferable to point estimates that look empirical. Either way, parameter mappings should be pre-registered before outcome comparisons are seen, using the same freezing mechanism this study applies to its confirmatory protocol.

Three model changes are prerequisites rather than refinements, and are named in the specification: layered recovery endpoints over 72 hours, 21 days, 90 days and a 180-day tail; nonzero residual failure for isolated backups; and either multi-facility structure or an explicit narrowing of the paper's scope to a single synthetic facility.

7 Conclusion

Placeholder section.

Research integrity and declarations

Ethics. The study used synthetic networks, simulated trajectories, public organization-level incident records, a public vulnerability catalog, and published aggregate findings. It involved no participants, patient-level records, private organizational data, malware, or intervention. No external interviews were conducted. Consent, privacy, and institutional requirements must be reviewed before any future human-participant or organizational data collection.

Data and code. Code, configuration, raw simulation outputs, public source snapshots with hashes, processed tables, figures, tests, frozen protocols, and run manifests are at <https://github.com/Dodgeboi/NazeResearchNSRI>. A submitted version should cite a fixed release or archived commit. **No restricted Medicare claims data are included;** the only hospital-linked file is a public replication deposit.

Artificial-intelligence use. Claude, ChatGPT, and Codex were used for coding assistance, debugging, literature-search terms, analysis checks, organization, and editing. An AI system also performed the audit that identified the defects described in the introduction and implemented the corrections, under human direction. AI responses were not treated as evidence. Numerical claims are generated from committed data by script, and literature claims were checked against cited sources. The authors remain responsible for the analysis and the wording. No AI system is an author.

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